

How do we encounter rare factor deficiencies in children? Single-centre results from Turkey

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Background Rare factor deficiencies (RFDs) are autosomal recessively inherited coagulation factor deficiencies encountered at a frequency of between one in 500 000 and one in two million.

Materials and methods One hundred and ninety-two patients, diagnosed as having RFD, followed and treated in our clinic between 1990 and 2013 were retrospectively evaluated in this study.

Results From the 192 patients, 142 had FVII, 15 had FX, 14 had FXI, 10 had fibrinogen, six had FV, two had FXIII, two had FV + FVIII and one had FII deficiency. One hundred and thirty of the cases were boys and 62 were girls. The age range was 2 weeks to 24 years and the ages at the admission were between 2 weeks and 16 years. The rate of consanguinity was 49.4%. Eighty-eight of our patients were asymptomatic (45.8%) and 104 were symptomatic (54.2%). Asymptomatic patients were diagnosed by family histories (39.8%), preoperative laboratory studies (54.6%) and operational bleeding (5.6%). Sixty-eight of our symptomatic patients displayed grade II (65.4%) and 36 displayed grade III bleeding symptoms (34.6%). First bleeding regions were skin (33%), nose (28%), central nervous system (CNS) (15.5%), oral cavity (10.5%), soft tissue (6%), joint (3%), urinary system (2%) and gastrointestinal system (GIS) (2%), respectively. The bleeding prevalence rates of our symptomatic patients are listed as epistaxis 62.5%, skin bleedings 53%, oral cavity bleeding 28.8%, haematomas 18.3%, CNS bleedings 17.3%, haemarthrosis 14.4%, GIS

bleedings 3.8%, menorrhagia 2.9%, haematuria 1.9%, bleeding because of operations 1.9% and iliopsoas bleedings 1.9%. CNS bleedings (41%) take the first place among the serious bleedings of our cases, followed by haemarthrosis (36.4%), GIS bleedings (18.1%) and iliopsoas bleedings (4.5%). Prophylaxis was applied to nine patients (five patients with FVII, two patients with fibrinogen and one each with FV and FX deficiency).

Conclusions The characteristics of clinical presentations, first bleeding attacks, bleeding prevalence and severe bleedings as well as prophylactic approaches are discussed in this article. *Blood Coagul Fibrinolysis* 26:145–151 Copyright © 2015 Wolters Kluwer Health, Inc. All rights reserved.

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Keywords: childhood, clinical findings, prophylaxis, rare factor deficiency, severe bleeding

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Introduction

Rare factor deficiencies (RFDs) include deficiencies of fibrinogen, FII, FV, FV + FVIII, FVII, FX, FXI FXIII and vitamin K-associated factors. They constitute 3–5% of all inherited factor deficiencies [1]. They are more frequently seen in the Middle Eastern countries including our country, Iran and India, where consanguineous marriages are common. It is also reported that they are encountered more frequently than they are known to be experienced in this geography [2,3].

Although patients with RFDs may not display any symptoms, they may also reveal themselves with serious life-threatening bleedings. Because of the high frequency rates of asymptomatic patients as in our country, clinicians may encounter difficulties for the clinical follow-up. Generally, RFDs display milder symptoms compared with haemophilia cases. FX deficiency may resemble

haemophilia A and B with serious bleedings. Afibrinogenemia, FVII and FXIII deficiency cases may also display serious bleedings. They may cause gynaecological bleedings and obstetric problems particularly in female patients, umbilical cord bleedings during the neonatal period and central nervous system (CNS) bleedings [4–7].

RFD patients classified two groups according to factor levels and functions. Patients with type I deficiencies have low factor coagulant activity (F:C) levels and type II patients have normal and even high factor antigen levels, but factor's functions are defective. Another difference between the patients with RFD and haemophilia is the types of mutations. Although intron 22 and intron 1 inversion are responsible for 50% of the cases in haemophilia A, mutations are almost specific to people with RFDs [8]. The relationship between the factor activities

and the clinical phenotype was researched in many studies. In contrast to the haemophilia patients, factor activities and the severity of clinical symptoms are not consistent with each other in RFD. A recent study has reported consistency between fibrinogen, FX and FXIII activities and the clinical phenotype, whereas a weak association for FV and FVII and no association at all for FXI [9]. RFD cases are patients that require care and attention in terms of both their diagnoses and treatments.

Many international and national centres and study groups are trying to compile information related to RFDs. The World Federation of Haemophilia (www.wfh.org) and the Rare Bleeding Disorders Database (www.rbdd.org) are the most important ones among these groups [8,10]. There is no such programme in our country. Asymptomatic cases might be diagnosed during trauma, circumcision, surgical interventions, pregnancy or menstruation. There are also problems for follow-up and insufficiencies the evaluation of the need of prophylaxis in RBD patients with severe bleeding episodes such as CNS, GIS and iliopsoas bleeding.

Clinical applications and observations, first bleeding regions, severe bleeding episodes and prevalence of bleeding as well as prophylactic characteristics of 192 RFD cases followed at our clinic since 1990 until today are reported in our study.

Materials and methods

One hundred and ninety-two RFD cases followed and treated in our clinic between 1990 and 2013 were retrospectively evaluated. Information of patients was retrieved from patient files and from the records contained in the electronic information processing environment created after 2005. Patient characteristics as well as their family histories, ages at presentation, first bleeding symptoms and bleeding episodes, transfusion requirements, prophylactic treatments and how asymptomatic cases were identified were recorded. Diagnostic tests of the patients such as blood count, prothrombin time (PT), activated partial thromboplastin time (aPTT), fibrinogen, bleeding time and blood smear were performed at the laboratory of our hospital. The factor activity and inhibitor determinations of the patients were worked on at the haemostasis laboratory of our hospital and the Internal Haematology Laboratory of the Cerrahpasa Medical Faculty of the Istanbul University. Patients' F:C and inhibitor assays were studied in our haemostasis laboratory and in Istanbul University, Cerrahpasa Medical Faculty, Internal Haemostasis Laboratory. Commercial kits were used for PT and aPTT tests. Fibrinogen concentration determinations were done by a modified Clauss method using a commercial reagent (Multifibren U; Dade-Behring/Siemens Healthcare Diagnostics Products GmbH, Marburg, Germany) and factor assays were performed as one-stage clotting-based tests using commercial plasma deficient of relevant factor on different

coagulation analysers (BCT, BCS of Dade-Behring/Siemens or CA-1500 of Sysmex) depending on availability. FXIII level was evaluated qualitatively by screening for lysis of formed clots in 5 mol/l urea, and when the test was positive, it was assumed that the FXIII level was below 5%. F:C Ag tests and mutation analyses were excluded. F:C Ag tests and mutation analyses could not be performed. Patients with factor activities lower than 50% were included in the study. For fibrinogen, noncoagulation was defined as afibrinogenaemia and the values under 100 mg/dl were defined as hypofibrinogenaemia. The classification recommended by the European Network of Rare Bleeding Disorders (EN-RBD) group was taken into consideration in the evaluation of clinical phenotypes of the patients [9]. Patients are categorized into four groups according to clinical bleeding severity. In asymptomatic patients, there were no documented episodes of bleeding. Grade I bleeding was defined as bleeding that occurred after trauma or drug ingestion (antiplatelet or anticoagulant therapy). Grade II bleeding was defined as spontaneous minor bleeding (bruising, ecchymosis, minor wounds, oral cavity bleeding, epistaxis and menorrhagia). Grade III bleeding was defined as spontaneous major bleeding: haematomas (intramuscular requiring hospitalization), haemarthrosis, CNS, gastrointestinal and umbilical cord bleeding. Written informed consent was provided by the families of the patients included in the study. Simple descriptive analyses, including mean, median and range, were used to summarize the results using SPSS v.10.0 for Windows (SPSS Inc, Chicago, Illinois, USA). Fresh frozen plasma (FFP), fibrinogen, rFVIIa, activated prothrombin complex concentrate and tranexamic acid were applied as antifibrinolytic treatment in the bleedings of the patients. All of the treatments were applied from peripheral veins. Catheters were not used. Inhibitor developments as well as thrombotic and allergic problems were recorded during treatments and subsequent observation.

Results

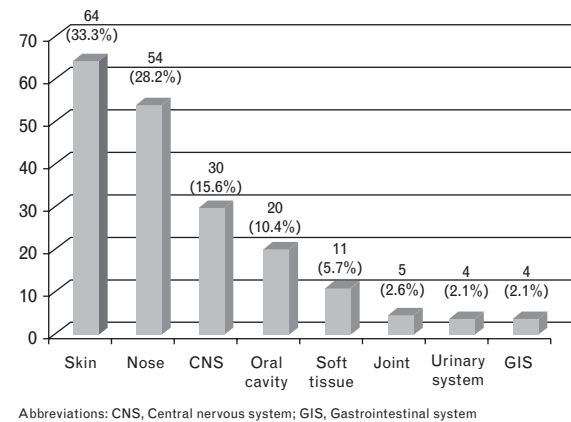
In this study, 192 RFD patients were assessed between 1990 and 2013. One hundred and forty-two of the cases had FVII (74%), 15 had FX (7.8%), 14 had FXI (7.3%), 10 had fibrinogen (5.3%), six had FV (3.1%), two had FV + FVIII (1%), two had FXIII (1%) and one had FII (0.5%) deficiency. One hundred and thirty (67.7%) of the cases were boys and 62 (32.3%) were girls. Their ages ranged between 2 weeks and 24 years (13 ± 5.27 years) and their age at first bleeding ranged from 2 weeks to 16 years (5.4 ± 4.5 years). The proportion of consanguinity was 49.4%. Among these, 50% of patients had a first-degree relationship, 35% of patients had second degree and 15% of patients had a third-degree relationship. The F:C levels of patients were less than 5% in 48 patients, between 5 and 30% in 40 patients and between 30 and 50% in 104 patients. Eighty-eight of our patients were

asymptomatic (45.8%) and 104 were symptomatic (54.2%). Asymptomatic patients were diagnosed by family histories (39.8%), preoperative laboratory studies (54.6%) and operational bleeding (5.6%). Sixty-eight of our symptomatic patients displayed grade II (65.4%) and 36 displayed grade III bleeding symptoms (34.6%). The ages at the admission of symptomatic patients were between 2 weeks and 16 years (1.12 ± 0.8 years). Forty-two of symptomatic cases were diagnosed before the age of 1 year (40.4%). Genetic analyses were performed to only eight patients with FVII deficiency.

Prophylaxis was applied to nine patients (five patients with FVII deficiency, two patients with afibrinogenemia and one each with FV and FX deficiency). Development of inhibitors and thrombotic incidents was not observed during bleedings and prophylaxis treatments. An allergic reaction to FFP developed in one FV deficiency case. None of our patients was lost. The distribution of the patients according to their factor deficiencies, sex, first bleeding time, consanguinity, F:C activities, clinical phenotype and prophylactic treatments applied to the them as well as their transfusion requirements have been shown in Table 1.

First bleeding regions were skin in 64 patients (33.3%), nose in 54 patients (28.2%), CNS in 30 patients (15.6%), oral cavity in 20 patients (10.4%), soft tissue in 11 patients (5.7%), joint in five patients (2.6%), urinary system in four patients (2.1%), and gastrointestinal system (GIS) in four patients (2.1%). First bleeding regions of our patients have been indicated in Fig. 1.

Fig. 1



First bleeding regions of the patients. CNS, central nervous system; GIS, gastrointestinal system.

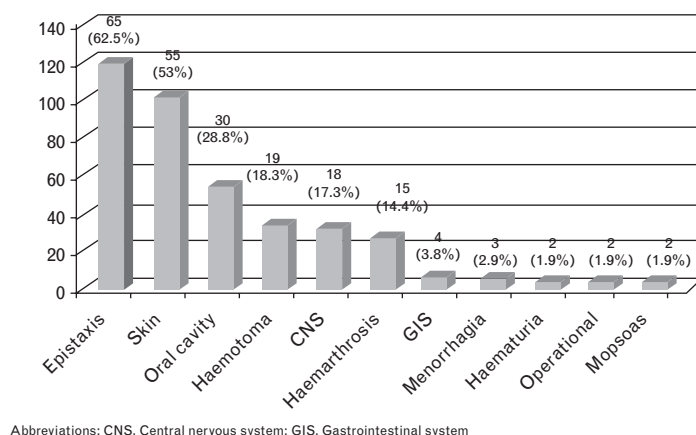
The bleeding prevalence rates of our symptomatic cases are listed as epistaxis in 65 patients (62.5%), skin bleedings in 55 patients (53%), oral cavity bleeding in 30 patients (28.8%), haematomas in 19 patients (18.3%), CNS bleedings in 18 patients (17.3%), haemarthrosis in 15 patients (14.4%), GIS bleedings in four patients (3.8%), menorrhagia in three patients (2.9%), haematuria in two patients (1.9%), bleeding owing to operations in two patients (1.9%) and iliopsoas bleedings in two patients (1.9%). The bleeding prevalence rates of the patients are seen in Fig. 2.

Table 1 Demographic features of the patients ($n = 192$)

	Fibrinogen deficiency	Factor II deficiency	Factor V deficiency	Factor V + VIII deficiency	Factor VII deficiency	Factor X deficiency	Factor XI deficiency	Factor XIII deficiency	Overall, n , (%)
Patients (n)	10	1	6	2	142	15	14	2	192
Sex(n)									
Male	6	–	5	–	98	10	9	2	130 (67.7)
Female	4	1	1	–	44	5	5	–	62 (32.3)
First bleeding time (n)									
<3 month	1	–	2	–	9	7	–	–	19 (18.3)
3 month–1 year	3	–	–	–	13	4	1	2	23 (22.1)
1–5 year	3	–	1	1	19	2	4	–	30 (28.8)
>5 year	1	1	1	–	28	–	1	–	32 (30.8)
Consanguinity (n)	7	–	3	–	64	13	7	1	95 (49.5)
F:C (n)									
<5%	6	1	2	–	20	9	8	2	48 (25)
5–30%	2	–	4	2	26	5	1	–	40 (20.8)
30–50	2	–	–	–	96	1	5	–	104 (54.2)
Clinical phenotype (n)									
Symptomatic	8	1	4	1	69	13	6	2	104 (54.2)
Grade I	–	–	–	–	–	–	–	–	–
Grade II	5	1	2	–	49	5	6	–	68 (65.4)
Grade III	3	–	2	1	20	8	–	2	36 (34.6)
Asymptomatic	2	–	2	1	73	2	8	–	88 (45.8)
Family history	1	–	1	1	26	2	4	–	35 (39.8)
Preoperative test	1	–	–	–	44	–	3	–	48 (54.6)
Operational bleeding	–	–	1	–	3	–	1	–	5 (5.6)
Transfusion need (n)	1	–	2	1	10	6	1	1	22 (21.2)
Prophylaxis (n)	2	–	1	–	5	1	–	–	9 (4.7)

F:C, factor coagulant activity.

Fig. 2



Abbreviations: CNS, Central nervous system; GIS, Gastrointestinal system

The bleeding prevalence rates of the patients. CNS, central nervous system; GIS, gastrointestinal system.

CNS bleedings (41%) take the first place among the serious bleedings of our cases, followed by haemarthrosis (36.4%), GIS bleedings (18.1%) and iliopsoas bleedings (4.5%). The distribution of severe bleedings according to bleeding regions has been shown in Table 2. The distribution of severe bleedings according to ages has been shown in Fig. 3. The distribution of patients with severe bleeding according to factor activity has been shown in Table 3.

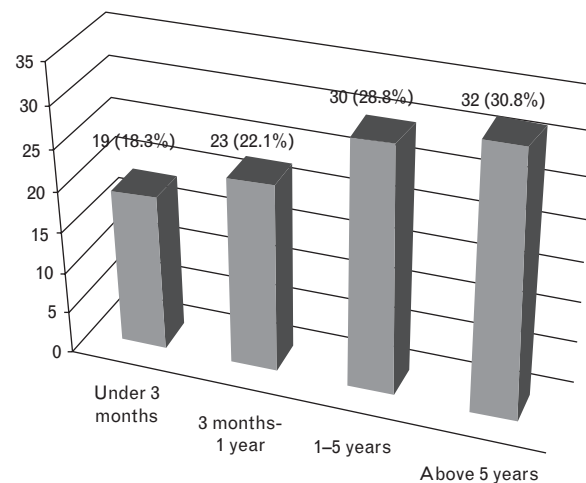
Discussion

RFD is not rare in our country, where consanguineous marriages are common. We do not have a national registration system covering these patients. According to 2010 data of our Ministry of Health, 398 cases that applied for medicine cards have been registered [10]. Two hundred and fifty-seven (64.6%) of the patients have FVII, 74 (18.6%) have FX, 40 (10%) have FXIII, 20 (5%) have FV and seven (1.8%) have FXI deficiency. These figures do not cover unreported heterozygote cases and borderline deficiencies. We do not have the epidemiological data of registered patients, their bleeding frequencies, and morbidity and mortality results. FVII cases are the largest

group within our RFDs (74%). FX cases take the second place with 7.8%. The World Federation of Haemophilia reported 16 075 RFDs from 105 countries in 2011. Of the patients, 32% had FVII and 29% had FXI deficiency [11]. According to EN-RBD data, including 489 patients, FVII and FXI deficiencies seem to be the most prevalent RBDs worldwide, with a frequency of 28% and 22% of the total RBDs, respectively [8]. It was known that distribution of RFD cases exhibits geographic and ethnic differences [12–14]. Also patients may be undiagnosed or misdiagnosed. In this respect, it may be appropriate to determine the priorities of each country according to its own data [15].

Rare deficiencies of clotting factors are quite common in our country and there is an increasing number of

Fig. 3



The distribution of symptomatic patients with severe bleedings according to ages ($n = 104$).

Table 2 The distribution of severe bleedings according to bleeding regions

	CNS	Haemarthrosis	Gastrointestinal	Iliopsoas	Overall
Fibrinogen	1	2	1	–	4
Factor II	–	–	–	–	–
Factor V	1	–	–	–	1
Factor V + factor VIII	1	–	–	–	1
Factor VII	10	9	7	–	26
Factor X	5	3	–	1	9
Factor XI	–	–	–	–	–
Factor XIII	–	2	–	1	3
Overall (%)	18 (41%)	16 (36.3%)	8 (18.2%)	2 (4.5%)	44

CNS, central nervous system.

Table 3 Patients with severe bleeding distribution according to factor activity

	F:C <1%	F:C 1–5%	F:C 5–30%	F:C 30–50%	Overall
Fibrinogen	4	–	–	–	4
FII	–	–	–	–	–
FV	1	–	–	–	1
FV + FVIII	–	–	1	–	1
FVII	8	5	7	6	26
FX	4	4	–	1	9
FXI	–	–	–	–	–
FXIII	3	–	–	–	3
Overall (%)	20 (45.5%)	9 (20.4%)	8 (18.1%)	7 (16%)	44

F:C, factor coagulant activity.

publications on this subject [16,17]. Our clinic admits patients from every region of our country and our RFD cases diagnosed for several reasons are increasing. Even though the rates of girls and boys are expected to be close in RFD cases, higher rate of boys (67.7%) among our patients suggests that the complaints of boys might be considered more important. Some of the patients are diagnosed during their medical examinations prior to circumcision. Presenting age of patients ranges between 2 weeks and 16 years (1.12 ± 0.8 years). We think that low mean age of patients at presentation may cause low percentage of menorrhagia. Although the amount of literature about children in RFD is really low, our group completely reflects the children's age group. A high rate of consanguinity (49.4%) exists in the history of our patients. It is known that autosomal recessive disorders including RFD are more frequently seen in countries where consanguineous marriages constitute a sociocultural phenomenon [2,3,18]. The ages of diagnosis were taken into consideration; 59.6% were understood to have been diagnosed over the age of 1 year. Early diagnosis constitutes a problem in our country. We have also another problem to look at the genetic mutations. Genetic analysis was performed in only eight patients with FVII deficiency. T359M mutation was detected in two patients with moderate and severe phenotype. G2835 mutation was in two patients with severe phenotype. R353Q mutation with mild phenotype in one and heterozygote R353Q mutation with moderate phenotype in one. T359M heterozygote mutation with moderate phenotype in one and T324M/R353Q mutation with mild phenotype in one patient (16). We have enough cases to start to look at the genetic mutations and determine the founder effects of our cases at least for Turkey and we are planning to start the genetic analysis.

In RFD patients, factor activity does not seem to be consistent with bleeding patterns [19]. For this reason, patients with RFD are classified according to the clinical phenotype, not according to factor activities such as the ones who have haemophilia. The first classification suggestion was made by IRF7 (International Registry on Congenital FVII Deficiency Database) for patients with FVII deficiency [20]. Also EN-RBD group has been

suggested a similar classification for patients with RFD [9]. We also used this classification for our patients who are categorized in four groups according to clinical bleeding severity. Grade I bleeding is defined as bleeding that occurred after trauma or drug ingestion. There is no case with grade I bleeding in our patient group. It can be considered redefinition of grade I in children.

Asymptomatic patients comprise an important portion of RFD cases in our patient group (45.8%) and they are most frequently diagnosed during the preoperative stage (54.6%). These are followed by family history (39.8%) and bleedings (5.6%) encountered during surgical intervention. EN-RBD group has reported 224 (45.8) asymptomatic patients among the 489 patients. They reported grade III, II, I bleeding in 61 (12.5%), 117 (23.9%) and 87 (17.8%) patients, respectively [8]. Thirty-six of our symptomatic patients displayed grade III (18.75%) and 68 displayed grade II (35.4%) bleeding symptoms.

Skin bleedings (33.3%) and epistaxis (28.2%) stand out as first bleeding symptoms for our symptomatic patients. These are followed by CNS bleeding (15.6%), oral cavity bleedings (10.4%), haematomas (5.7%), haemarthrosis (2.6%), haematuria (2.1%) and GIS bleedings (2.1%).

In symptomatic patients, first bleeding times are under the age of 3 months in 19 (18.3%) patients, between 3 months and 1 year in 23 (22.1%) patients, between 1 and 5 years in 30 (28.8%) patients and above the age of 5 years in 32 (30.8%) patients. Early presentation ages were observed in FX and F cases. Seven of 13 symptomatic FX patients (53.8%) were diagnosed within the first 3 months and 11 of them (84.6%) were diagnosed within the first year. Two of four symptomatic FV patients were also diagnosed within the first 3 months. FX and FV patients were presented in early ages with bleedings. The rate of FVII deficiency in patients presented with bleeding within the first 3 months was 13%. In symptomatic patients with FVII deficiency, clinical practice seems to spread over time.

The bleeding prevalence rates of our RFD cases are listed as epistaxis 62.5%, skin bleedings 53%, oral cavity bleedings 28.6%, haematomas 18.2%, CNS bleedings 17.2%, haemarthrosis 14.6%, GIS bleedings 3.7%, menorrhagia 3.1%, haematuria 1.9%, bleeding owing to operations 1.9% and iliopsoas bleeding 1.9%. The most common bleeding regions were skin and mucous membrane and this was consistent with the literature [1,21].

On the contrary, low rates of menorrhagia contrary to other publications arise from the young ages of the patients. Based on the information reported by the North America Rare Factor Deficiency Study Group, the most frequent bleedings according to their regions in RFD cases were in skin and mucosa (63%). The rate of CNS bleedings associated with FX deficiency was 15%. The rate of CNS bleedings was around 5–10% in fibrinogen,

FII, FV, FVII and FXIII deficiencies. No symptoms were observed in 40% of homozygote FVII cases [4]. In 2004, Manucci *et al.* [22] reported the bleeding prevalence rates for 750 Iranian patients as: epistaxis in 66%, menorrhagia in 65%, haemarthrosis in 38%, postnatal and postoperative bleedings in 50%, oral cavity bleedings in 30% and CNS bleedings in 8% patients.

Among 44 severe bleeding episodes, CNS was the most common site ($n = 18$, 41%). It is followed by haemarthrosis ($n = 16$, 36.4%), GIS bleeding ($n = 8$, 18.1%) and iliopsoas bleeding ($n = 2$, 4.5%). The majority of our cases are FVII deficiencies and bleedings in FVII-deficient patients who make up the 55.9% of the severe bleedings. In 70.6% of our cases, the severe bleeding episodes were seen in the first 3 months of life. The rate of our patients with severe bleeding diagnosed during the first year reaches 92%. In 45.5% of patients with severe bleeding, factor activity is lower than 1%. However, it is realized that in 34.1% of patients with severe bleeding, factor activity is also 5–50%. This result is especially prominent in patients with FVII deficiency. Factor activity is 1–5% in 15 patients and 5–50% in 13 patients, with FVII deficiency and severe bleeding. On the contrary, severe bleeding was detected in eight of nine patients with FX deficiency and their factor activity was less than 5%. Peyvandi *et al.* [9] reported that there was a heterogeneous association between coagulation factor activity level and clinical bleeding severity in different RBDs and they observed a strong association only in fibrinogen, FX and FXIII deficiencies. Our results are consistent with the literature.

Ragni *et al.* [21] reported in 1981 that CNS bleeding was observed in 12 of the 75 FVII deficiency cases. Peyvandi and Mannucci [22] reported CNS bleedings as 17% among the bleeding frequencies concerning 49 patients and Mariani and the IRF7 study group reported CNS rate as 7% (24/21). Herrmann *et al.* [23] reported that although CNS rates were 2% in homozygote cases, no CNS bleeding was seen in heterozygote cases. Karimi *et al.* reported CNS bleedings in an RFD group consisting of 126 patients as 10 (7.9%). Five of these patients had FVII, three had FXIII and two had FX deficiency [24].

These results support the fact that severe bleedings reveal themselves at early ages in RFD cases and emphasize the importance of early diagnosis. Diagnosing and treating severe bleedings during the neonatal and early infancy periods will be a factor that will decrease mortality and morbidity. Rare deficiencies of clotting factors must be kept in mind in differential diagnosis of bleeding episodes in neonates and infants. CNS bleedings are different though. In 66.6% of the patients, the CNS bleeding is the first bleeding episode. This emphasizes the importance of CNS bleedings. We think that CNS bleeding cases could be identified during early

stages because of the close observation of the neonatal units of our hospital.

Transfusion was required to be applied to a total of 22 patients during the bleeding treatments of our RFD cases (21.2%). It was stated during a study that anaemia was discovered in 19–49% of the cases and transfusion was required in 10–20% [4]. One of our significant deficiencies is genetic studies. Our eight FVII deficiency patients, who were subjected to mutation-type analyses, were reported earlier [16].

FFP, PCC, activated prothrombin complex concentrate and rFVIIa were used in the treatment of our patients. The preparations of FX, FXI and FXIII that were not available in our country should be obtained. Allergic reactions to 2–26% plasma-based products were reported in homozygote FX, FV, FVII and FXIII deficiency patients. Anaphylaxis was observed at the rates of 2–5% against plasma products in FVII and FX deficiency cases [4]. Allergic reaction was observed in one of our FV deficiency cases, who was given prophylaxis with FFP because of intracranial bleeding and such allergy was taken under control with antiallergic drugs. No inhibitor development or thrombosis was observed during the bleeding attacks and prophylactic treatment processes of our cases. None of our patients were lost.

The need of prophylaxis in RFD is emphasized occasionally [25–27]. Publications about the prophylaxis were as case reports exception of FXIII and FVII deficiency [27–30]. The information we have about prophylaxis is either limited, short-termed or only covers the period of a surgical intervention. We believe that prophylaxis is an option that should be evaluated in patients, who apply with severe bleeding. In our series, five patients with FVII deficiency were given prophylaxis of rFVIIa weekly, two patients with FV deficiency were given FFP weekly, two patients with fibrinogen deficiency were given fibrinogen every 2 weeks and one patient with FX deficiency was given PCC twice a week. Our patients are receiving prophylaxis for periods ranging between 10 months and 7 years. There are no determined standard prophylaxis programmes for RFD patients. It is recommended in life-threatening bleedings or those that could result in disabilities. Prophylactic experiences particularly for FVII, FX and FV deficiency should be evaluated on a patient-by-patient basis.

RFDs are clinically not well defined and their treatments are not specified clearly, in contrast to more frequent bleeding disorders. Both before and during the diagnosis, these patients may go through considerable clinical problems and cause diagnostic and therapeutic challenges and an economic burden that differs for each country. Increasing numbers of record trials for each individual factor deficiency or as a whole are being conducted in many countries in recent years. There is no such trial in Turkey yet.

Acknowledgements

Conflicts of interest

There are no conflicts of interest.

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